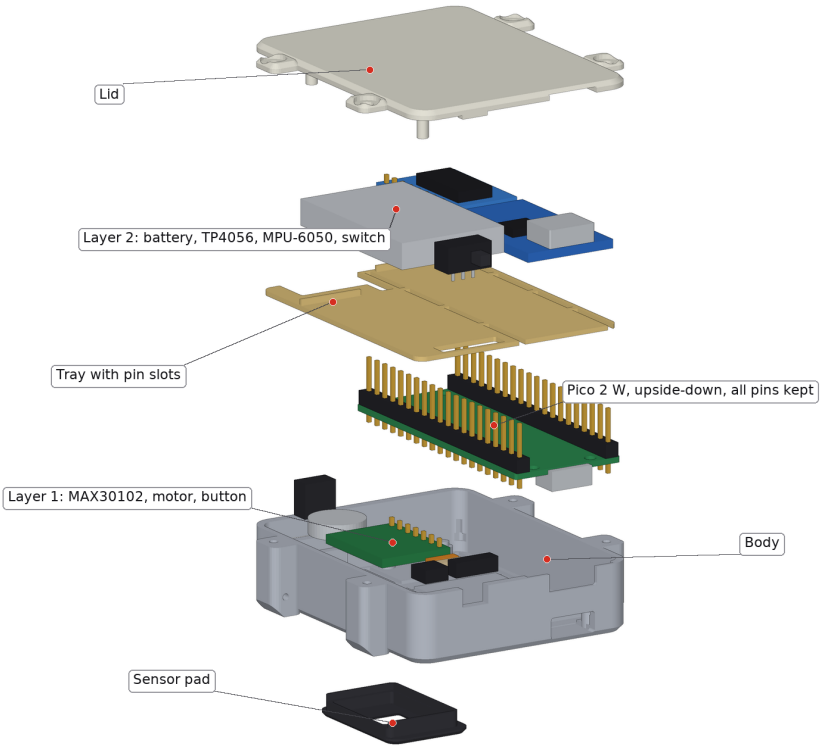
Sanad — wrist band body v2b

3D-print files, wiring plan and step-by-step assembly: from the printed parts to wearing it

Body size	43 mm across the wrist × 54.6 mm along the forearm × 14.0 mm tall (+ 1 mm sensor bump underneath)
Strap	Any 22 mm quick-release watch strap (lug-to-lug 53 mm)
Printed parts	4: body, sensor pad, tray, lid — no supports needed
Inside	Raspberry Pi Pico 2 W with every pin kept, MAX30102, MPU-6050, TP4056 USB-C charger, 402030PT LiPo, push button, slide switch, coin vibration motor

What is new in v2b. The Pico goes in **upside-down**, so its header pins are never cut and the board still works in other projects. The long pins rise through slots in the tray and stop in pockets in the lid. The TP4056 and MPU-6050 sit between the two pin rows.

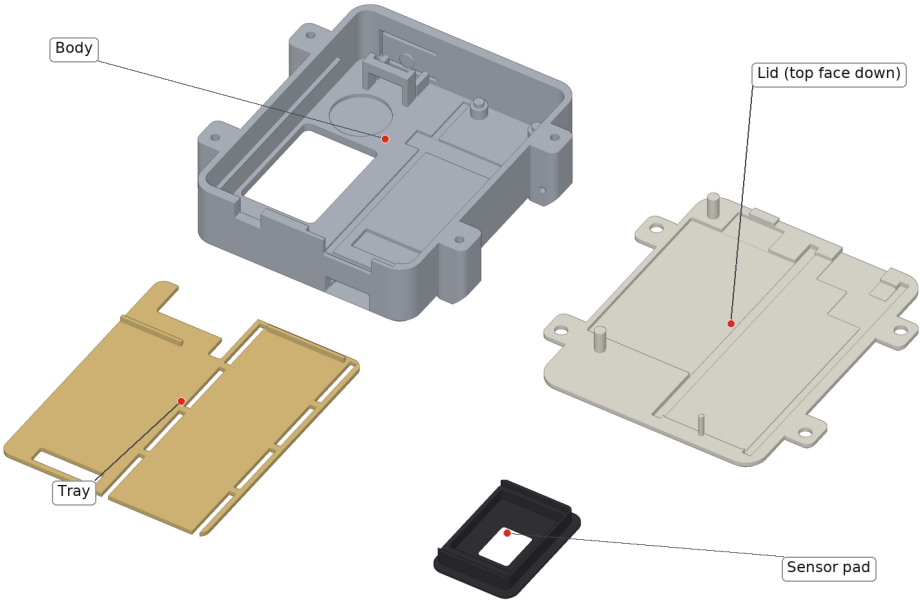
Do not print yet. Several sizes are still estimates (section 3). The TP4056 must be 17.1 mm wide or less, because it sits between the pin rows. The design has been checked in the 3D model only; nothing has been printed or test-fitted.



1. Ordering the print

Once the sizes in section 3 are confirmed, you can send the files straight to an online 3D-printing service. If the service asks for units, choose millimetres. The files are in **models/**.

File	What it is	Qty	Sits on the bed	Colour
sanad-body.stl	Main case with lugs	1	bottom down (as worn)	any
sanad-sensor-pad.stl	Bump that holds the heart-rate sensor	1	bump down	black
sanad-tray.stl	Shelf between the two layers, with pin slots	1	flat, ribs up	any
sanad-lid.stl	Lid (already flipped for printing)	1	top face down, pegs up	white / natural



The four parts in their print orientation, rendered from the files.

Settings to ask for

Setting	Value	Why
Material	PETG (best) or PLA	PETG handles body heat and sweat and lets the button tab flex without cracking. Resin or nylon (MJF) also work well.
Layer height	0.12–0.16 mm	small holes, thin tabs and the pin slots come out cleaner
Walls / perimeters	3 (≈1.2 mm)	the side walls are 1.2 mm, so they print solid
Infill	40 %	
Supports	None	every part was designed to print without them
Lid colour	white or natural	the charger LEDs glow through a 0.6 mm thin window in the lid
Sensor-pad colour	black	keeps stray light out of the heart-rate sensor

2. What you need

Part	Notes	Qty
Raspberry Pi Pico 2 W	headers fitted; **the pins stay as they are**	1
MPU-6050 (GY-521)		1
MAX30102 module (green, 7 pins)		1
TP4056 Type-C charger with protection	Makers, 18 EGP. **Must be 17.1 mm wide or less.** Charge-current change needed, see Step 8	1
LiPo 402030PT, 350 mAh real, with protection	Makers, 65 EGP	1
Mini push button 2-pin 6×6×5 mm	Makers, 1 EGP (buy 3)	1
SK12D07VG3 right-angle slide switch	Makers, 3.5 EGP	1
RCD1027 coin vibration motor (10 × 2.7 mm)	Makers, 15 EGP	1
1N5817 Schottky diode	battery → VSYS	1
2N2222A or S8050 NPN transistor (TO-92)	motor driver	1
1N4148 diode	across the motor	1
1 kΩ resistor	transistor base	1
4.7 kΩ resistor, 0603 or 0805 SMD	replaces the TP4056 charge-current resistor	1
M2 × 6 mm pan-head self-tapping screws	lid	4
22 mm quick-release watch strap	any brand	1
30 AWG silicone wire (4–5 colours)	about 1 m total; outer diameter 0.8 mm or less	—
Kapton tape, thin double-sided tape (0.4 mm), heat-shrink		—

Tools

- Soldering iron with a fine tip, solder, flux, and desoldering wick for the TP4056 resistor swap
- Flush cutters (for the MAX30102 and MPU-6050 only), tweezers, small pliers, a hot-glue gun, and super glue (CA)
- Drill bits 1.2 mm and 1.6 mm in a pin vise, for cleaning holes by hand
- A breadboard to hold the Pico while you solder
- Small Phillips screwdriver, multimeter, calipers, and safety glasses — snipped pins fly

3. The estimated sizes

These are the sizes no reliable source gives for your parts, so the model uses estimates. Measure them before you print. “Largest that fits” is the limit found by growing one part at a time inside the printed case.

Part	Measure	Model uses	Largest that fits
Pico header	PCB to long pin tip, plastic included	8.5 mm	8.75 mm
	Black plastic strip height	2.5 mm	2.6 mm
	Short pin end past the PCB	2.0 mm	2.55 mm
MAX30102	Board length	18.1 mm	19.2 mm
	Board width	13.6 mm	14.0 mm
	Board thickness	1.6 mm	sets heights
	Left (VIN) edge to the left end of the chip	7.7 mm	window moves to match
	Pin-row edge to the near side of the chip	4.45 mm	window moves to match
	Total thickness, board + chip	3.15 mm	sets heights
	Total thickness, board + tallest capacitor	3.5 mm	3.7 mm
MPU-6050	Board thickness	1.6 mm	plenty of room
	Total thickness, board + tallest part	3.6 mm	4.5 mm
TP4056	**Width (decides whether this layout works)**	17.0 mm	**17.1 mm**
	Length	29.0 mm	29.6 mm
	Total height, PCB bottom to USB-C top	5.2 mm	5.35 mm
	USB-C sticks out past the board edge	0.6 mm	sets its position
Battery	Width	20.5 mm	20.5 mm
	Length with its protection board	32.5 mm	32.7 mm
	Thickness at the thickest point	4.0 mm	4.0 mm on 0.4 mm tape
Switch	Knob length, leg spacing (not published)	3.0 mm, 2.0 mm	tell the designer

Hold the MAX30102 sensor side up, pin row towards you, VIN on the left. Also check two things by eye: the long Pico pins are on the plain side of the board, and whether the board has a small 3-pin white debug connector, which is not in the model.

> **Why the TP4056 width matters.** It sits between the pin rows, which are only 17.14 mm apart. Makers lists it at 28 × 17 mm, which leaves about 0.07 mm each side. Other sellers list 17.3 to 17.4 mm, and those will not fit. If yours is wider than 17.1 mm, sand the board edges a little, or use the older layout in **archive/v2-flipped-pico**, which puts the charger on the floor.

These sizes are already known, so skip them: the Pico board and micro-USB, the MAX30102 chip, the slide switch body, the button and the motor (Appendix A).

```
PICO: long pins on plain side= debug connector= PCB-to-tip= plastic= short end=
MAX: L= W= t= VINedge-to-chip= pinrow-to-chip= total+chip= total+cap=
MPU: t= total+tallest=
TP4056: L= W= total@USBC= USBC sticks out=
BATTERY: L= W= T=
SWITCH: knob length= leg spacing=
```

4. Wiring plan

The two sensors are on separate I²C buses, as in v1, so the code needs no pin changes. **All Pico wires go on the short pin ends**, on the parts side. The long pins never get solder on them.

The Pico is upside-down, so pins 1–20 face the sensor side and pins 21–40 run along the wall. Wires from pins 31–39 cross under the board, in channels in the floor, and come out on the sensor side.

From	To (Pico pin)	Wire
Battery red / black	TP4056 B+ / B−	≈35 mm, between the pins and over the MPU-6050
TP4056 OUT+	Switch middle leg	≈50 mm, down the switch slot
Switch outer leg	1N5817 end **without** the stripe	≈10 mm
1N5817 stripe end	39 VSYS	≈25 mm, under the Pico
TP4056 OUT−	38 GND	≈60 mm, down the switch slot and under the Pico
MAX30102 VIN / GND	36 3V3(OUT) / 8 GND	≈55 / 10 mm
MAX30102 SDA / SCL / INT	6 GP4 / 7 GP5 / 9 GP6 (INT optional)	≈15 mm
MPU-6050 VCC / GND	36 3V3(OUT) / 33 GND	≈70 / 60 mm, under the Pico and up the opening
MPU-6050 SDA / SCL / INT	31 GP26 / 32 GP27 / 34 GP28 (INT optional)	≈50–65 mm, the same way
Button legs	19 GP14 and 18 GND	≈10 / 25 mm
GP15 → 1 kΩ → transistor B	20 GP15	≈50 mm
Transistor E / C	13 GND / motor blue −	≈40 / 45 mm
Motor red +	MAX30102 VIN pin (3V3)	≈10 mm. 1N4148 across the motor, stripe to red +

Lengths are along the model’s routes. Add about 15 mm to each; you can shorten them during assembly. Use one colour per signal: red 3V3, black GND, yellow SDA, blue SCL.

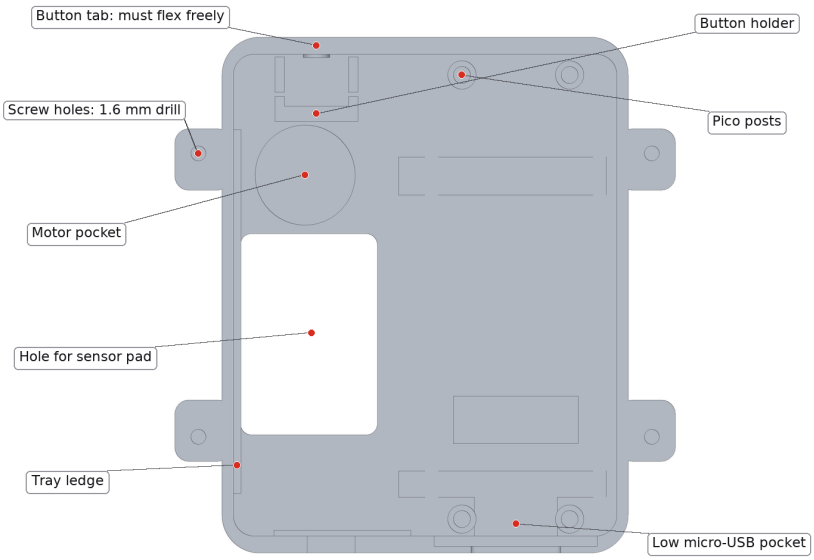
The battery reaches VSYS through a Schottky diode, as Raspberry Pi’s Pico 2 W datasheet describes for a single lithium cell. The diode also stops the Pico’s own USB from back-feeding the battery when you plug it in to program it. Charging only happens through the TP4056’s USB-C.

5. Assembly, step by step

Work in this order: **the case first, then layer 1 on the floor, then the tray, then layer 2, then the lid.** Test before you close anything (Steps 6 and 9).

Step 1 — Clean up and check the print

1. Remove any strings or brim. Run a **1.2 mm** drill by hand into the four spring-bar holes (on the inner faces of the lugs) and a **1.6 mm** drill into the four screw holes on top of the lugs.
2. Press the button tab on the side wall: it must flex a little and spring back. If a thin skin closed one of its U-slots, open it with a craft knife.
3. Check the four Pico posts on the floor are complete, and that the floor hole for the sensor pad is clean.



The body seen from above (button side at the top, ports side at the bottom).

Step 2 — Fit the sensor pad

1. Push the black sensor pad into the floor hole **from the outside**, bump facing out, until its rim sits flat against the bottom of the case. Its three low fences end up inside the case.
2. Fix it with a few drops of super glue around the plug, inside the case. Keep glue out of the window.

Step 3 — Pin headers: the Pico keeps its pins

1. **Pico 2 W: do nothing.** Leave every pin and both black plastic strips as they are.
2. MAX30102 and MPU-6050 only: wear safety glasses, then prise the black plastic strip up and off the pins with a flat screwdriver, or nibble it apart with flush cutters.
3. Snip their pins. On the side that will get wires, leave a **1.5 mm** stub. On the other side, snip **flush**.
4. Re-touch every joint on those two boards with the iron afterwards, because snipping can crack them.

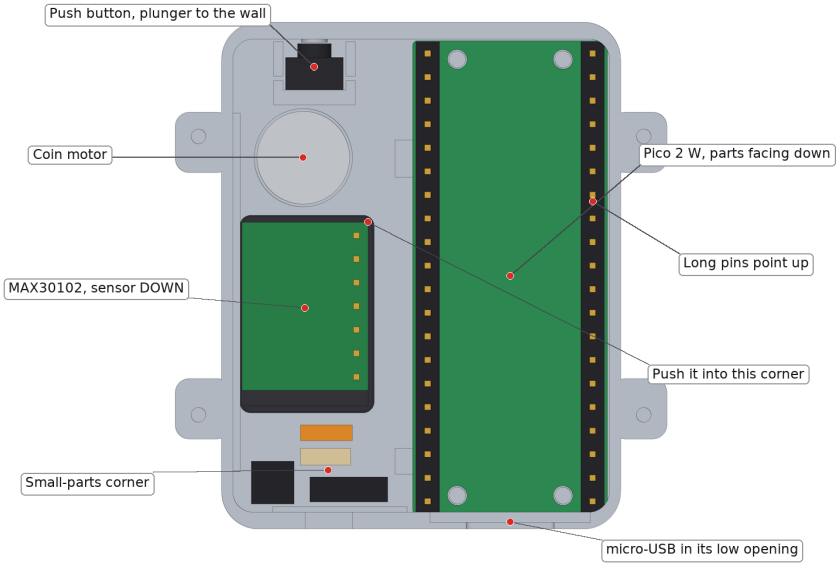
Board	Wire side	Other side
Pico 2 W	short pin ends, nothing cut	long pins and plastic strip stay on
MAX30102	back (the plain side), 1.5 mm stubs	**sensor side: must be flush**, it faces the pad
MPU-6050	top, 1.5 mm stubs (wires arrive from above)	bottom, flush

Step 4 — Pre-wire the boards

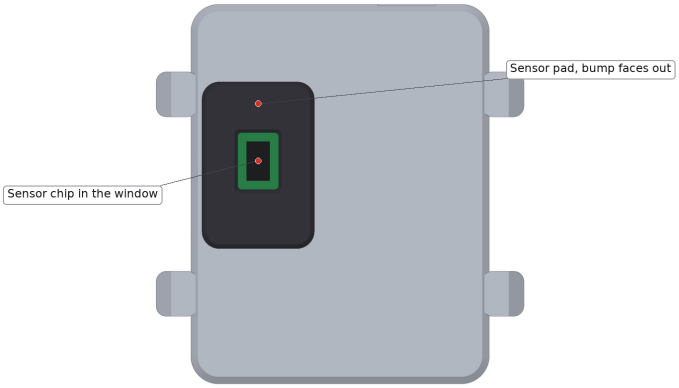
1. Flash MicroPython onto the Pico first: BOOTSEL faces the floor once it’s fitted.
2. Solder wires to each board using the table on the wiring page. On the Pico, use the **short pin ends** only, low down, with the long pins pushed into a breadboard to hold it.
3. Run the wires from pins 31–39 across the parts side, between the pins, and tape them flat with Kapton, so all 15 wires leave along the pin 1–20 edge.
4. Slide heat-shrink onto any splice before you solder it.

Step 5 — Layer 1: Pico, sensor, motor, button

1. **Pico 2 W: turn it over, parts facing down.** The long pins now point up, into the case.
2. USB toward the ports wall: tilt it, put the micro-USB into its low opening, then lower the far end onto the four posts.
3. **MAX30102**, **sensor face DOWN**, pin row toward the Pico, **VIN end toward the button side**. Push it into that corner of the fences and check from underneath that the dark sensor chip sits in the window. Fix the board edges with hot glue; never on the chip.
4. **Coin motor** into its round pocket, wires up, held with a drop of hot glue.
5. **Push button** into its holder with the plunger against the wall, touching the nub on the button tab, and its two legs pointing left and right. Hot-glue it.



Layer 1 from above. The Pico’s plain side faces you.



From underneath: the sensor chip must sit in the window. If it doesn’t, slide the board until it does before the glue sets (the window has 1.3 mm of margin each way).

Step 6 — Small parts, then a first test

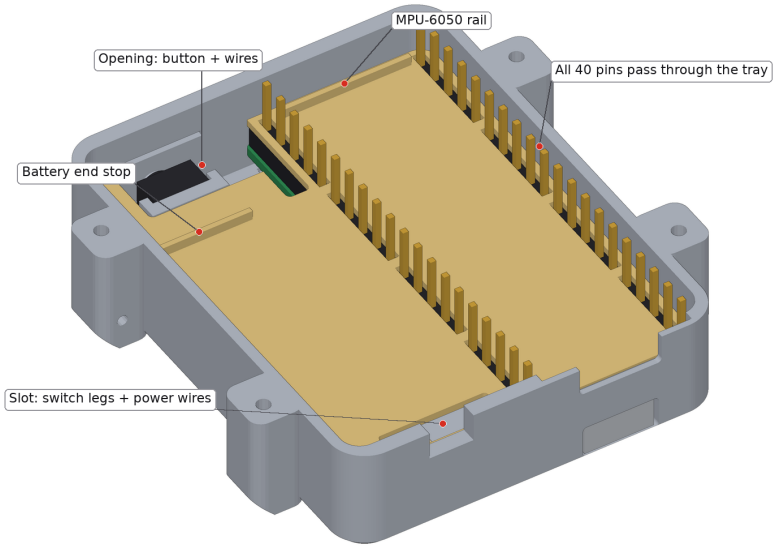
1. Put the 1N5817, the transistor with its 1 kΩ resistor, and the 1N4148 in the small-parts corner on the floor. Cover every bare lead with heat-shrink or Kapton.
2. Solder the layer-1 wires: MAX30102 to I²C0, button to GP14, the motor driver to GP15, and the 1N5817 to VSYS.
3. Plug the Pico into your computer with micro-USB (battery not connected yet) and run this in MicroPython:

```
from machine import Pin, I2C
import time
i2c0 = I2C(0, sda=Pin(4), scl=Pin(5), freq=100000)    # MAX30102
i2c1 = I2C(1, sda=Pin(26), scl=Pin(27), freq=100000)  # MPU-6050
print('I2C0', [hex(a) for a in i2c0.scan()])   # expect ['0x57']
print('I2C1', [hex(a) for a in i2c1.scan()])   # expect ['0x68'] (needs the MPU wired)
motor = Pin(15, Pin.OUT); motor.value(1); time.sleep(0.4); motor.value(0)
btn = Pin(14, Pin.IN, Pin.PULL_UP)
while True:
    print('pressed' if btn.value() == 0 else '-'); time.sleep(0.3)
```

Press the button tab on the outside of the case. The output should change to **pressed**.

Step 7 — Tray

1. Pass the wires that go up to layer 2 through the tray: the MPU-6050 wires through the opening near the button, and the power wires through the slot under the switch.
2. Lower the tray **straight down over all 40 pins**, onto its ledge and the plastic strips. Line up both rows first and don’t force it. The top of the button passes through the tray’s opening.

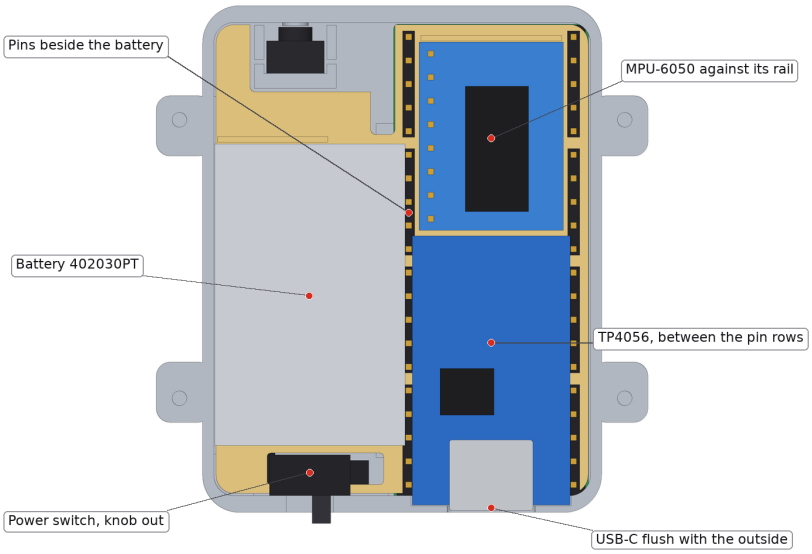


Tray in place. The pins pass through its slots.

Step 8 — Layer 2: charger, battery, switch, MPU-6050

Do this before the first charge. TP4056 boards usually ship set to about 1 A (1.2 kΩ, marked “122”). That is about 3× the capacity of a 350 mAh cell. The TP4056 datasheet table gives 300 mA at 4 kΩ and 250 mA at 5 kΩ. Replace the resistor next to the PROG pin (pin 2) with a **4.7 kΩ** one for about 250 mA. Until you’ve done it, don’t charge this battery.

1. Swap the PROG resistor on the bench: 1.2 kΩ (“122”) out, 4.7 kΩ (“472”) in. It’s harder to reach between the pins.
2. **TP4056**, components up, USB-C toward the ports wall: a strip of Kapton on the tray, then lower it **straight down between the two pin rows**. The board end drops into the wall and the USB-C face sits flush with the outside. Check no pin gets bent, then hot glue at the far end.
3. **Battery**, with Kapton along the long edge facing the pins, on thin tape (0.4 mm if you can) against its end stop, wires toward the button side. Never pierce or bend the cell.
4. **Power switch**: knob out through its slot in the ports wall, legs down through the tray slot. Hot-glue it.
5. **MPU-6050** on thin tape between the pin rows, against its rail. Keep it in this direction so its axes match the code.



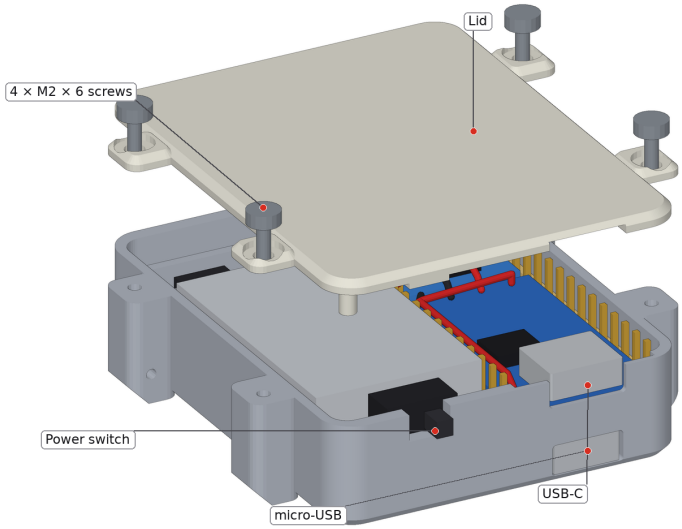
Layer 2 from above.

Step 9 — Power test

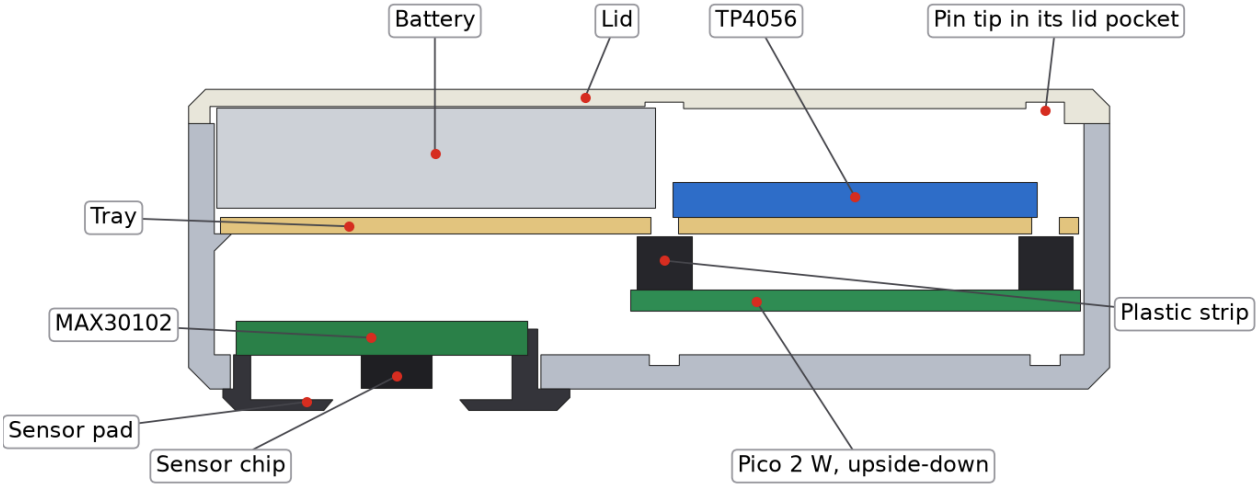
1. Solder the battery to B+ / B−, with its wires crossing the near pin row and running over the MPU-6050. Switch the power switch ON. VSYS (pin 39) should read the battery voltage minus about 0.3 V, and the Pico should boot.
2. Plug a USB-C charger in. The red charging LED should glow through the lid's light window once the lid is on, and a different colour shows when the battery is full.
3. Switch OFF and check that VSYS drops to 0 V. The battery still charges with the switch off.

Step 10 — Close the lid

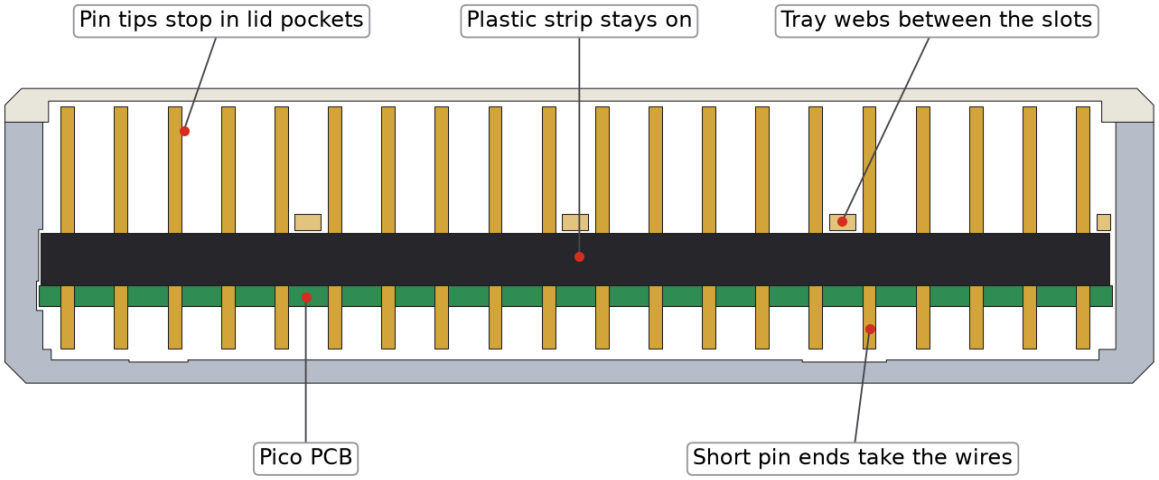
1. Tuck the wires flat so nothing presses on the battery. Place the lid: the pin tips go into its two long pockets, the thin light window goes over the TP4056, and the small tab fills the top of the switch slot. Three pegs on the lid hold the tray down, and that holds the Pico.
2. Drive in the four M2 × 6 screws by hand until they're snug. Don't over-tighten; they cut their own thread in the plastic. If the lid won't sit flat, stop and look for a pin or wire in the way.



Lid and screws.



Cross-section through the sensor, cut from the real model.



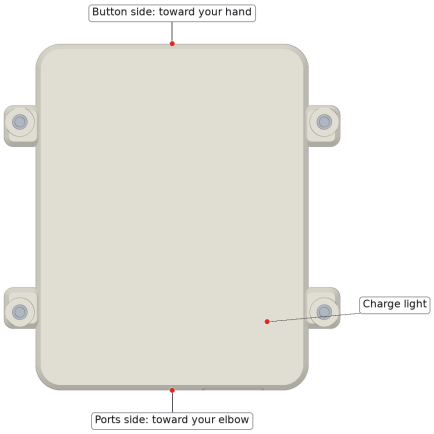
Cross-section along the pin row nearest the battery. The pins stay whole from end to end.

Step 11 — Fit the strap

1. Slide a spring bar into the end of one strap half. Put one tip into a lug hole, pull the quick-release lever (or press the other tip with a pin) and let it snap into the opposite hole.
2. Tug it firmly. If a tip is loose, the hole printed too big: a drop of super glue inside the hole, left to dry, tightens it.

Step 12 — Wear it

- Wear it on top of the wrist, just above the wrist bone. Point the **button side toward your hand** and the **ports toward your elbow**, so the button is where a watch crown would be.
- Tighten the strap until the sensor bump is pressed against the skin but you can still slide one finger under it. The heart-rate reading needs that contact.
- Charge through the USB-C on the ports side without taking the strap off. To program it, take the band off, because the micro-USB sits low.



Top view as worn.

Getting the Pico back for another project

Take off the lid, lift out the tray with layer 2, lift out the Pico, then unsolder the wires from the short pin ends. The long pins are clean and ready for a breadboard.

Appendix A — Dimensions taken from real sources

Dimension	Value	Source
Pico length, width, PCB	51 × 21 × 1 mm	Raspberry Pi Pico 2 W datasheet, Fig. 3
Pico hole Ø, spacing, centre to end	2.1 mm, 11.4 mm, 2 mm	Pico 2 W datasheet, Fig. 3
Pico pin pitch, row spacing	2.54 mm, 17.78 mm	Pico 2 W datasheet, Fig. 3
micro-USB width, overhang	8 mm, 1.3 mm	Pico 2 W datasheet, Fig. 3
micro-USB height above PCB	2.73 mm	Official Raspberry Pi 3D STEP model
micro-USB sunk into the board edge	0.77 mm	Official Raspberry Pi 3D STEP model
MAX30102 chip	5.6 × 3.3 × 1.55 mm	Maxim MAX30102 datasheet
MPU-6050 board	20.26 × 15.56 mm	components101 drawing on the Makers product page
TP4056 Type-C board	28 × 17 mm	Makers Electronics listing (other sellers list up to 17.4 mm wide)
Battery cell	4 × 20 × 30 mm	Makers Electronics listing, 402030PT
Motor	Ø10 × 2.7 mm	Makers Electronics listing, RCD1027
Button	6 × 6 × 5 mm	Makers Electronics listing
Switch body	8.7 × 4.4 × 4.7 mm	LCSC and Makers listings, SK12D07VG3
Spring-bar tip, body	Ø0.9 mm, Ø1.8 mm	WatchGecko and Archer quick-release spring bars
M2 pan head	Ø4 mm, 1.6 mm high	ISO 7049 / ISO 7045

Design choices: 1.2 mm side walls, 1.8 mm end walls, 1.6 mm floor and lid, 0.6 mm thinnest lid skin, 0.8 mm tray, 1.0 mm sensor bump, 4 mm outer corners, and 1.25 mm spring-bar holes.

Appendix B — Troubleshooting

Problem	Fix
A part doesn’t fit its pocket	Don’t force it. Put the real size in MEASURE at the top of cad/build.py and rebuild; the model is fully parametric.
The tray won’t go over the pins	A pin is bent. Straighten it gently with tweezers. Never file or cut it.
The TP4056 won’t go between the pins	Measure it. If it’s wider than 17.1 mm, sand the board edges a little, or use the older layout with the charger on the floor.
I²C scan doesn’t show 0x57	Check SDA and SCL aren’t swapped. Some MAX30102 boards pull the I²C lines up to 1.8 V only; if yours does, add 4.7 kΩ pull-ups to 3V3.
Heart-rate signal weak	Tighten the strap a notch, make sure the chip is centred in the window, and raise the LED current in software.
Button doesn’t click	The nub must touch the plunger. Add a 0.5 mm shim (tape) behind the button.
Lid won’t close	Usually a wire under the battery or a pin tip missing its pocket. Check both.
Weak Wi-Fi	The Pico’s antenna faces the wrist with the MPU-6050 above it. This hasn’t been tested yet, so check the signal early.
Spring bar falls out	See Step 11, or use spring bars with longer tips.
Battery gets warm while charging	Unplug it and check the charge-current resistor (Step 8).